

Day 4 : Types of Cyber Attacks & Detection Fundamentals

Objective

- Develop a clear understanding of common cyber attack techniques
- Identify key indicators associated with each attack type
- Understand the role of a SOC Analyst in detecting and analyzing threats
- Apply basic detection techniques using logs and behavioral patterns

1. Phishing Attack

Definition

Phishing is a **social engineering attack** in which attackers impersonate legitimate entities to deceive users into revealing sensitive information such as login credentials, OTPs, or financial details.

Common Characteristics

- Emails or messages appearing to originate from trusted organizations
- Use of urgent or alarming language (e.g., “*Immediate action required*”, “*Account will be suspended*”)
- Embedded links directing users to fake or spoofed websites
- Slight variations or misspellings in domain names

Attack Mechanism

1. Attacker crafts a deceptive message
2. User clicks on a malicious link
3. User enters sensitive information
4. Attacker captures and exploits the data

Prevention Techniques

- Verify sender email addresses carefully
- Avoid clicking on unknown or suspicious links
- Enable Multi-Factor Authentication (MFA)
- Educate users on identifying phishing attempts

2. Malware Attack

Definition

Malware refers to **malicious software** designed to infiltrate, damage, or gain unauthorized access to systems and data.

Types of Malware

Type	Description
Virus	Attaches to legitimate files and spreads when executed
Ransomware	Encrypts files and demands payment for decryption
Spyware	Monitors user activity and steals sensitive information

Infection Vectors

- Downloading untrusted or pirated software
- Opening malicious email attachments
- Visiting compromised websites

Real-World Scenario

A user installs unauthorized software → Malware executes in the background → Attacker gains remote access and control over the system

Prevention Techniques

- Install software only from trusted sources
- Use updated antivirus/endpoint protection tools
- Regularly update operating systems and applications
- Avoid opening unknown attachments

3. Brute Force Attack

Definition

A brute force attack is a method where attackers systematically attempt multiple password combinations until the correct one is identified.

Key Indicators

- Repeated login failures within a short time period
- Multiple login attempts from a single IP address
- Login attempts using common or weak password patterns

Attack Behavior

Attackers use automated tools to rapidly test large numbers of password combinations, targeting weak authentication mechanisms.

Prevention Techniques

- Enforce strong password policies
- Implement account lockout mechanisms
- Use CAPTCHA to prevent automated attempts
- Enable Multi-Factor Authentication (MFA)

SOC Detection Fundamentals

A Security Operations Center (SOC) is responsible for continuous monitoring, detection, and response to security incidents.

Detection Techniques

1. Log Monitoring

- Analyze authentication logs, system logs, and network logs
- Identify anomalies in user activity

2. Pattern Recognition

- Detect repeated login failures (brute force attempts)
- Identify unusual login times or locations

3. Threat Indicators

- Suspicious URLs or email patterns (phishing)
- Unexpected system behavior (malware infection)

SOC Detection Workflow

1. **Log Generation** – Systems generate logs
2. **Collection** – SIEM tools aggregate logs
3. **Alerting** – Suspicious activity triggers alerts
4. **Investigation** – Analysts validate the alert
5. **Response** – Appropriate mitigation actions are taken

Real SOC Scenario

Alert:

Multiple failed login attempts detected

Analyst Actions:

- Review source IP address
- Check login frequency and pattern
- Validate user activity

Response:

- Block suspicious IP
- Reset user credentials
- Escalate incident if required

Activity – Phishing Analysis

Scenario:

“Your account will be suspended. Click here immediately to verify your details.”

Analysis :

The given message is a **phishing attempt**.

Reasons:

1. Urgent and Threatening Language

The phrase “*account will be suspended*” creates fear and pressure.

- This is commonly used in phishing to force quick action without thinking.

2. Lack of Legitimate Source

The message does not mention:

- A verified sender
 - Official organization details
- Legitimate services clearly identify themselves.

3. Suspicious Call-to-Action

The instruction:

“Click here immediately”

- Indicates a malicious link that may redirect to a fake login page to steal user credentials.

Conclusion :

This message exhibits key phishing characteristics such as urgency, lack of authenticity, and a suspicious link.

Therefore, it should **not be trusted or interacted with**.

Recommended Action :

- Do not click the link
- Verify the message through official sources
- Report it as phishing

Assignment – Phishing Analysis

1. Real-World Phishing Example

A common real-world phishing example involves attackers impersonating trusted brands such as PayPal.

Example Email:

“Your PayPal account has been limited due to suspicious activity.
Please click the link below to verify your account immediately:
<http://paypal-verification-secure-login.com>”

2. Reasons Why This is Phishing

◆ Suspicious URL

The link does not belong to the official **PayPal** domain.

- Instead of paypal.com, it uses a fake domain designed to look legitimate.

◆ Urgent and Threatening Language

The message creates panic using phrases like:

“account has been limited” and *“verify immediately”*

- This pressures users to act without verifying authenticity.

◆ Request for Sensitive Information

The link redirects to a fake login page that asks for:

- Username
 - Password
 - Possibly OTP or banking details
- Legitimate companies never ask for sensitive details through suspicious links.

□ 3. Preventive Measures

◆ Verify URLs Carefully

- Always check the domain name before clicking
- Ensure it matches the official website (e.g., paypal.com)

◆ **Avoid Clicking Unknown Links**

- Do not click links from unexpected emails or messages
- Access accounts directly by typing the official URL

◆ **Enable Multi-Factor Authentication (MFA)**

- Adds an extra layer of security even if credentials are compromised

◆ **Check Email Authenticity**

- Look for spelling errors, unusual formatting, or unknown sender addresses

◆ **Use Security Tools**

- Enable spam filters and antivirus protection
- Keep systems updated

Conclusion

Phishing attacks rely on **deception and urgency** rather than technical complexity.

Awareness, careful verification, and basic security practices are essential to prevent such attacks.